

L13: Estimating the Number of Edges and the Average Degree

We can also apply the Horvitz-Thompson estimator to "totals" that are defined over all possible node pairs. Here, each node pair $(i, j) \in V \times V$ is associated with a value $y_{i,j}$, and we want to estimate the total value:

$$\tau = \sum_{(i,j) \in V \times V} y_{i,j}$$

For instance, if $y_{i,j}$ is one when the two nodes are connected and zero otherwise, the total τ is twice the number of edges in the population graph.

Another example: it could be that $y_{i,j}$ is one if the shortest path between the two nodes traverses a given node- k (and zero otherwise). Then, the value τ is related to the betweenness centrality of node- k .

Suppose that we have a sample S of n node pairs, with the corresponding values $y_{i,j}$ of the sampled node pairs. The Horvitz-Thompson estimator tells us that an unbiased estimator of τ is:

$$\hat{\tau} = \sum_{(i,j) \in S \times S} \frac{y_{i,j}}{\pi_{i,j}}$$

Let us first apply this framework in **estimating the number of edges in the population graph**. Suppose that we perform *induced subgraph sampling, starting with n nodes chosen without replacement*. Then, the probability of sampling a node pair (i,j) of the population graph is

$$\pi_{i,j} = \frac{\binom{n}{2}}{\binom{N}{2}} = \frac{n(n-1)}{N(N-1)}$$

So, our estimate for the number of edges in the population graph is:

$$\hat{\tau} = 1/2 \sum_{(i,j) \in S \times S} \frac{y_{i,j}}{\pi_{i,j}} = |E^*| \frac{N(N-1)}{n(n-1)}$$

which means that we just need to multiply the number of sampled edges $|E^*|$ by a correction factor (the inverse of the fraction of sampled node pairs).

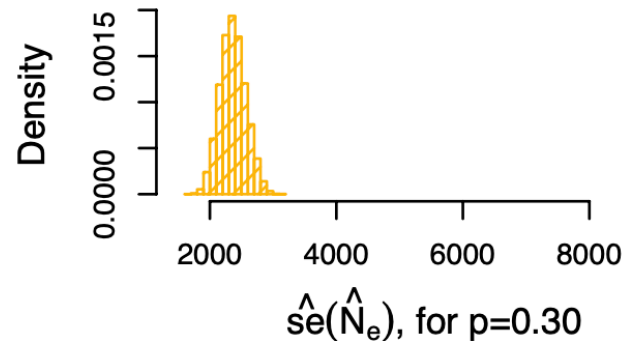
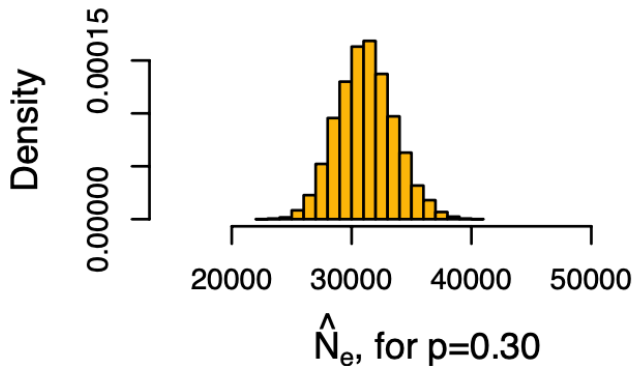
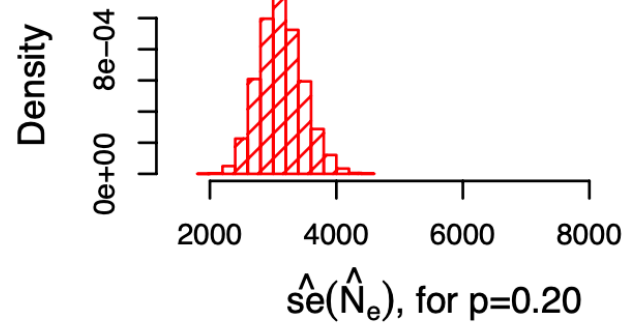
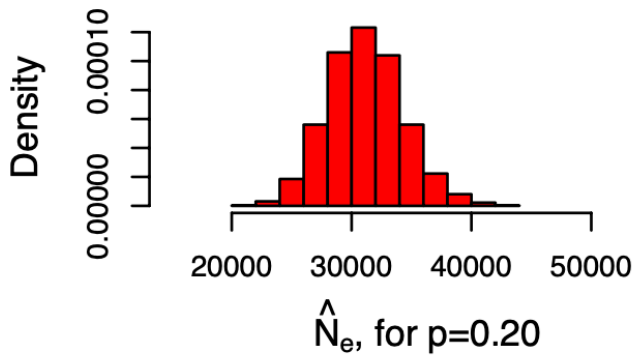
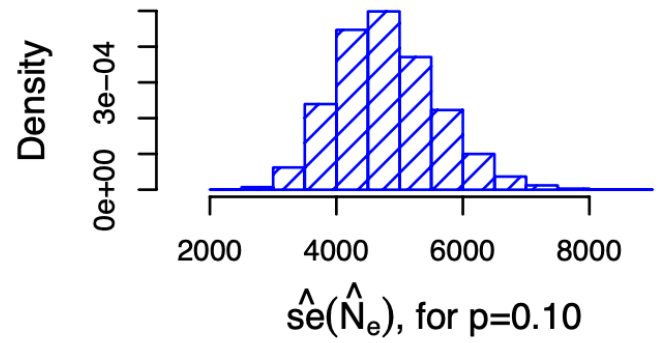
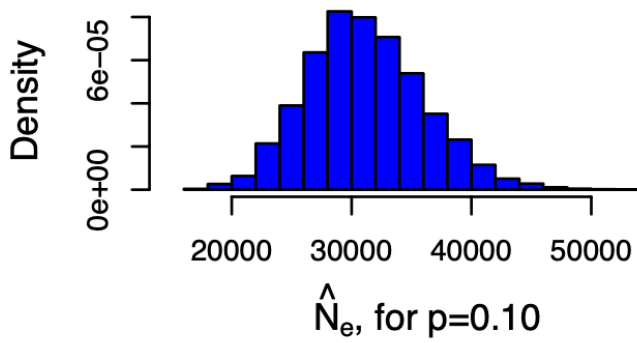


Image source: Kolaczyk, Eric D. Statistical Analysis of Network Data: Methods and Models (2009) Springer Science+Business Media LLC.

To put these results in a more empirical setting, the plot shows the results of estimating the network density (and thus the number of edges) in the *Yeast* protein-protein interaction network, when using induced subgraph sampling. The actual number of edges is 31,201 (and the number of nodes is 5,151). The plots at the left show the empirical distribution of the sampled number of edges $|E^*|$ for three node sampling fractions: $p=0.10$, 0.20 and 0.30 of the total number of nodes. The plots at the right show the empirical distribution of the standard error (estimate of standard deviation of $|E^*|$). All distributions are based on 10,000 trials.

Let us now use this result to also **estimate the average degree $E[k]$ in the population network**. Recall that the average degree is related to the number of nodes N and the number of edges $|E|$ as follows:

$$E[k] = \frac{2|E|}{N}$$

Using the previous estimate for the number of edges at the sampled graph, we get the following estimate for the average degree of the network:

$$\bar{k}_{\text{induced_subgraph}} = \frac{2}{N} |E^*| \frac{N(N-1)}{n(n-1)} = \frac{2 |E^*|}{n} \frac{N-1}{n-1}$$

It is interesting to compare this with the average degree estimate if we sample the network using **single-stage snowball sampling** (also known as **star sampling**). In that case, we sample all the neighboring edges of each sampled node. So, we know the exact degree k_i of each sampled node. Additionally, the inclusion probability for each node is n/N . So, the Horvitz-Thompson estimator for the total number of edges in the population graph is $\frac{1}{2} \sum_{i \in S} \frac{k_i}{n/N}$. Thus, the star-sampling estimate for the average degree is:

$$\bar{k}_{\text{star_sampling}} = \frac{2}{N} \left(\frac{1}{2} \sum_{i \in S} \frac{k_i}{n/N} \right) = \frac{2 |E^*|}{n}$$

Note that the induced-subgraph estimator differs from the star-sampling estimator by a factor $\frac{N-1}{n-1}$. That factor aims to correct for the extent to which the degree of each node is under-sampled when we use induced subgraph sampling.

Food For Thought

Recall that the Transitivity of a graph requires us to calculate the number of connected node triplets and triangles. How would you apply the previous framework to estimate these quantities from a sample of the graph, using induced subgraph sampling?